

Retrieval Augmented Generation

Haystack Dense Retrieval Practice Document

Introduction

Retrieval Augmented Generation (RAG) is a technique that combines information retrieval with generative language models. Instead of relying only on information stored in a model's parameters, a RAG system retrieves relevant external documents and provides them as context to a language model.

RAG Pipeline

A typical RAG pipeline contains document loading, text splitting, embedding generation, vector storage, retrieval, prompt construction, and answer generation. A user question is converted into a query representation, relevant passages are retrieved, and those passages are supplied to the language model.

Retrieval Methods

Keyword retrieval methods such as BM25 search for matching terms. Dense retrieval uses embeddings to represent questions and documents as vectors and finds semantically similar content. Hybrid retrieval can combine keyword and semantic approaches.

Advantages

RAG can improve factual grounding, make answers more relevant to private or changing documents, and allow knowledge to be updated without retraining the language model. Retrieved passages can also support citations and source attribution.

Limitations

RAG quality depends strongly on document quality, chunking, embeddings, retrieval accuracy, and prompt construction. If the retriever returns irrelevant passages, the generated answer may also be incorrect or incomplete.

This document is prepared as a small knowledge source for a Haystack BM25 versus dense retrieval experiment.